

Digital Marketing Customer Segmentation Dashboard

Google Analytics Sample Dataset | BigQuery | SQL | Looker Studio

Portfolio Project

Author - Hasna Kh

Project Overview

This project analyzes website traffic and customer behavior using the Google Analytics Sample Dataset in Google BigQuery. SQL was used to calculate key marketing metrics, including sessions, revenue, conversion rate, and average order value, while Looker Studio was used to build an interactive dashboard. The analysis provides insights into traffic sources, device performance, geographic trends, and overall business performance to support data-driven marketing decisions.

Technologies Used

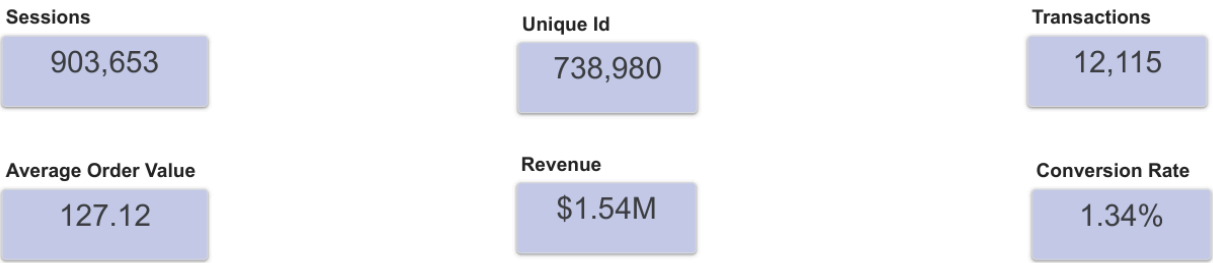
- Google BigQuery
- SQL
- Looker Studio
- Google Analytics Sample Dataset
- Git
- GitHub

Dataset

Source: Google Analytics Sample Dataset

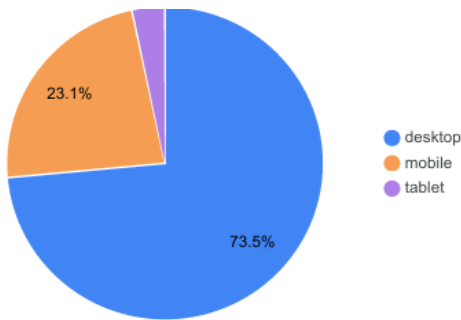
Table: `bigquery-public-data.google_analytics_sample.ga_sessions_*`

Overview



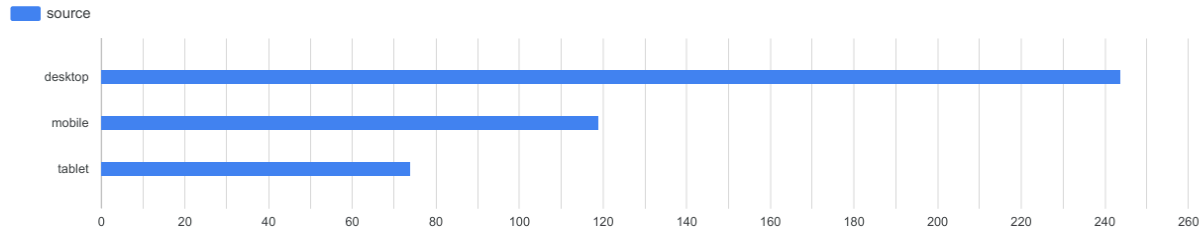
The Overview dashboard provides a high-level summary of website performance using key business metrics. It includes total sessions, unique visitors, transactions, revenue, average order value, and conversion rate. These KPIs help evaluate overall marketing effectiveness and customer activity at a glance.

Sessions by Device



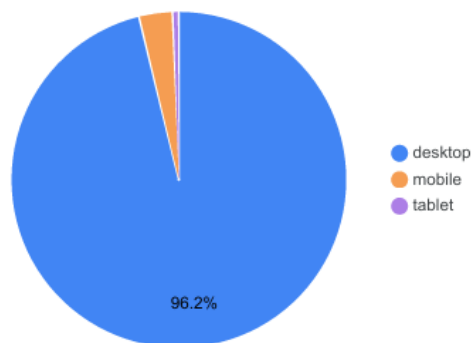
This chart shows website sessions across desktop, mobile, and tablet devices. Desktop users generated over 70% of total sessions, followed by mobile users, while tablet traffic was minimal.

Top Traffic Sources



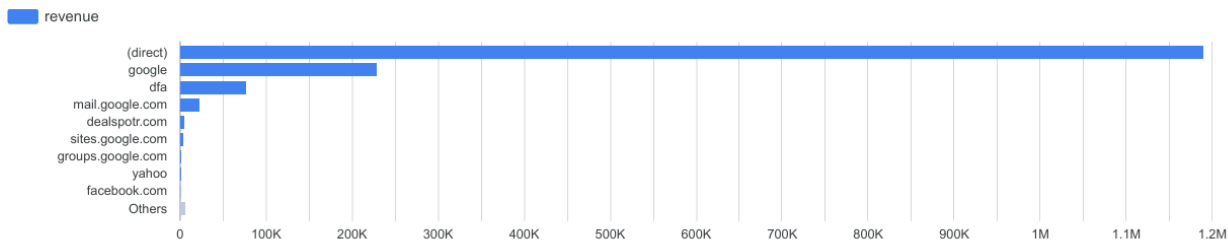
This visualization compares the number of sessions generated by different traffic sources. This visualization compares sessions across different traffic sources, helping identify the acquisition channels that generate the highest website traffic.

Revenue by Device



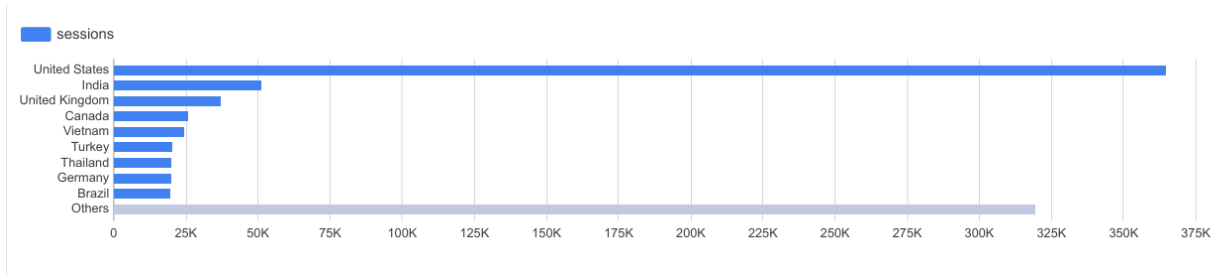
This chart illustrates the contribution of each device category to total revenue. Desktop users generate the majority of revenue, suggesting stronger purchasing behavior compared to mobile and tablet users.

Revenue by Source



This visualization highlights revenue generated from different traffic sources. Direct traffic contributes the highest revenue, followed by Google Search, indicating strong brand awareness and effective organic search performance.

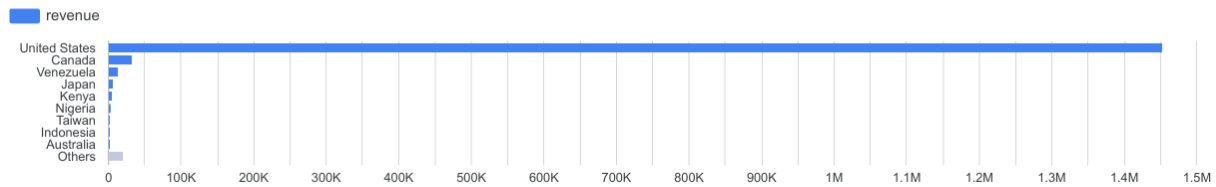
Top Countries by Sessions



This chart compares website sessions across different countries. The United States generates the largest share of traffic, followed by India, the United Kingdom, Canada, and several other regions. These insights help identify high-engagement geographic markets.

Revenue by Country

Revenue by Country



This chart shows revenue generated by users from different countries. The United States contributes the highest revenue by a significant margin, highlighting it as the most valuable market for business growth and marketing investment.

Key Insights

The dashboard summarizes the most important findings from the analysis:

- Desktop users generated over 70% of website sessions.
- Desktop devices contributed the majority of total revenue.
- Direct traffic produced the highest revenue among all acquisition channels.
- The United States was the leading country for both sessions and revenue.
- The average order value was approximately **\$127**, while the overall conversion rate was **1.34%**.

Conclusion

This project demonstrates practical skills in SQL, data analysis, KPI reporting, dashboard design, and business intelligence using Google BigQuery and Looker Studio. It highlights the ability to transform raw website data into actionable insights, making it a strong portfolio project for Data Analyst, Marketing Analyst, and Digital Marketing Analyst roles.

Recommendations

- Improve the mobile user experience to increase conversions.
- Invest further in high-performing acquisition channels such as Google Search.
- Prioritize marketing campaigns in high-value markets like the United States.
- Optimize the conversion funnel to improve the current 1.34% conversion rate.